Databases & warehousing

The revision notebook

36 lessons / 12 modules / 120 questions / 60 flashcards

Use this concise companion after studying the interactive lessons. Retrieve the rule, work a fresh example, and explain why an alternative answer fails. Full explanations and runnable examples are in the printable HTML notes and practice ZIP.

Scope: every named topic in Section 5 of the official GATE DA 2027 syllabus. Practical extensions are identified in the lessons. Python and DSA remain available as separate tracks in the same site.

Verified 24 October 2026. The 22 official PYQ entries link to the original papers and answer keys; they are references, not reproduced question papers.

Open the official syllabus

Your route through the unit

01. Model the data

Entities, relationships, keys and integrity before you write SQL

02. Relational algebra

Build queries from sets, joins, division and equivalences

03. Tuple relational calculus

Predicates, quantifiers, safety and translation into SQL

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Filtering, NULL, grouping, joins and subqueries on real tables

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Measurement scales, missing values, encoding and leakage

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Representative samples, streaming selection and compact representations

12. Warehousing and OLAP

Grain, star schemas, hierarchies, cubes and correct measure roll-ups

Rules to retrieve without notes

Concept	Recall rule
Keys	Superkey: determines every attribute. Candidate key: inclusion-minimal superkey. Prime: belongs to at least one candidate key.
SQL logic	WHERE keeps TRUE only. COUNT(*) counts rows; COUNT(x) ignores NULL. NOT IN with NULL can produce UNKNOWN.
Division	No required value is missing. Universal queries over an empty required set accept every candidate in the stated domain.
Closure	Start with X; repeatedly apply every FD whose left side is contained in the closure until no change.
Binary lossless test	R1 intersect R2 must determine all of R1 or all of R2 under F+.
3NF	For every nontrivial X->A, X is a superkey or A is prime.
BCNF	For every nontrivial X->A, X is a superkey. Lossless decomposition may lose dependency preservation.
Blocks	Unspanned bfr=floor(B/R); blocks=ceil(N/bfr). Subtract specified overhead before using usable B.
B+ internal fanout	pP+(p-1)K <= B. Leaf entries have a separate capacity equation.
Min-max	a+(x-min)*(b-a)/(max-min). Fit on training data; out-of-range future values can exceed [a,b].
Z-score	(x-mean)/sd. State population/sample deviation. This does not make a distribution Gaussian.
Cuboids	Base/ALL for d dimensions: 2^d. With Li allowed levels including ALL: product of Li.
Average roll-up	Combine sums and counts, then divide. Do not average unequal-size group averages.
Measure classes	Additive/semi-additive/nonadditive describe summation across dimensions. Distributive/algebraic/holistic describe required aggregate state.

CHAPTER 01

Model the data

Entities, relationships, keys and integrity before you write SQL

01 / From files to a relational database

A database stores related facts under a schema.

A relation has attributes with domains and tuples with one value per attribute.

A superkey uniquely identifies a tuple in every legal instance.

Begin a design by writing facts such as "one enrolment per student and course".

Watch the trap: A key is a constraint over all valid data, not an accidental property of the sample.

Retrieve by doing: Return the distinct cities in alphabetical order.

02 / Draw an ER model, then map it

An entity is a distinguishable object; an entity set groups objects of one kind.

A relationship associates entities and may have its own attributes: mark belongs to an enrolment, not to a student alone.

Map a strong entity to a relation with its key.

A weak entity depends on an owner and has a partial key that is unique only within that owner.

Watch the trap: A foreign key alone does not enforce that every parent has at least one child.

Retrieve by doing: Find courses with no enrolments; preserve the unmatched course.

03 / Integrity, domains and NULL

Domain constraints restrict values: type, allowed range, format or enumeration.

SQL expresses rules using PRIMARY KEY, UNIQUE, NOT NULL, CHECK and FOREIGN KEY.

NULL represents missing or inapplicable information, not zero, an empty string or a value equal to itself.

When a parent changes, referential actions may reject the operation, cascade it or set child keys to NULL, subject to the schema.

Watch the trap: WHERE score <> 80 also excludes NULL; it does not mean every row except the 80 row.

Retrieve by doing: List students with a missing score or a score at least 90.

CHAPTER 02

Relational algebra

Build queries from sets, joins, division and equivalences

01 / Selection, projection and set operations

Relational algebra is a compositional language: each operator takes relations and returns a relation.

Union, intersection and difference require union-compatible operands: the same degree and compatible corresponding domains.

Cartesian product pairs every tuple of R with every tuple of S.

To translate prose, identify the desired output attributes, the necessary source relations and predicates.

Watch the trap: Projection in relational algebra removes duplicates; a plain SQL SELECT does not.

Retrieve by doing: Use set difference to find all course IDs with no enrolment.

02 / Joins, multiplicities and query equivalence

A theta join selects pairs satisfying any stated predicate; an equijoin uses equality.

An inner join drops unmatched rows.

SQL bag joins multiply matching rows: if a key occurs a times on the left and b times on the right, it contributes a*b result rows.

Selections on one input commute with an inner join when their attributes belong to that input.

Watch the trap: ON and WHERE are not interchangeable for an outer join.

Retrieve by doing: Count each student's enrolments, including zero.

03 / Division and "for every" queries

Division answers a universal question.

Derive division by finding failures: candidates C = pi_A(R); required pairs = C x S; missing = (C x S) - R; bad candidates = pi_A(missing); result = C - bad candidates.

In SQL, "there is no required course that this student has not taken" becomes two nested NOT EXISTS predicates.

Universal quantification over an empty required set is true for every candidate (vacuous truth).

Watch the trap: Counting two enrolments does not prove both required courses were taken unless duplicates and irrelevant courses are handled.

Retrieve by doing: Find students enrolled in every course in the complete course table.

CHAPTER 03

Tuple relational calculus

Predicates, quantifiers, safety and translation into SQL

01 / Read and write tuple predicates

Tuple relational calculus (TRC) describes what should hold of the result, rather than an operator execution order.

A free variable is not bound by a quantifier and contributes to the answer.

For an output with selected fields, write the desired tuple of attributes explicitly, such as {<s.name> | Student(s) AND s.city="Delhi"}.

Translate a TRC formula by first listing free/output variables, then drawing quantifier scopes and finally enforcing membership in base relations.

Watch the trap: A variable bound inside EXISTS is not freely available as an output variable outside its scope.

Retrieve by doing: Return students with at least one enrolment below 70.

02 / Quantifier scope and universal conditions

EXISTS x P(x) needs one witness; FOR ALL x P(x) requires every candidate to satisfy the predicate.

Use FOR ALL x P(x) equivalent to NOT EXISTS x NOT P(x).

For "has no failing enrolment", use NOT EXISTS e (Enrol(e) AND e.sid=s.id AND e.mark<40).

Evaluate nested formulas from the inside out for a fixed outer candidate.

Watch the trap: "All marks are high" does not imply "there is a mark".

Retrieve by doing: Require at least one enrolment as well as no mark below 70.

03 / Safe calculus and expressive equivalence

A formula such as {t | NOT Student(t)} can range over infinitely many tuples outside the database.

Domain independence means the result does not change merely because the ambient universe contains additional unused values.

Safe relational calculus and relational algebra have equivalent expressive power for the classical relational operations.

To test a proposed answer, ask where every output value can come from, which variable is free, whether negation is guarded, and whether universal scope is correct.

Watch the trap: Safe negation starts with a finite candidate relation. An unconstrained complement can describe infinitely many tuples.

Retrieve by doing: Find courses not taken by student 3.

CHAPTER 04

SQL foundations

Filtering, NULL, grouping, joins and subqueries on real tables

01 / SELECT and the logical query pipeline

Read a query in logical order: FROM and joins construct rows; WHERE filters rows; GROUP BY forms groups; HAVING filters groups; SELECT constructs output; DISTINCT removes duplicate outputs; ORDER BY sorts; a row limit then restricts the display.

Use aliases to name relations and expressions, qualify ambiguous columns, and put text literals in single quotes.

ORDER BY can contain several expressions, each with its own ascending or descending direction.

SQL numeric types and conversion rules are dialect-sensitive.

Watch the trap: A SELECT alias is not universally available in WHERE; use the underlying expression or an outer query.

Retrieve by doing: Find the two highest nonmissing scores, breaking ties by student id.

02 / Aggregates, GROUP BY and HAVING

COUNT(*) counts rows; COUNT(x) counts non-NULL x values.

A GROUP BY creates one result per distinct grouping combination.

HAVING evaluates group-level predicates such as COUNT(*)>=2.

An overall average cannot generally be obtained by averaging group averages.

Watch the trap: NULL is ignored by AVG; it is not silently replaced with zero.

Retrieve by doing: Find cities whose average known score is at least 80.

03 / Subqueries, EXISTS and the NOT IN trap

A scalar subquery is intended to return one value; an IN subquery supplies a membership set; EXISTS tests whether any row is produced.

A correlated subquery refers to the current outer row.

NOT IN is dangerous when its list contains NULL.

ANY and ALL compare a value with a subquery's values: > ANY means greater than at least one; > ALL means greater than every one.

Watch the trap: If a subquery can contain NULL, NOT IN can reject every unmatched outer value.

Retrieve by doing: Express the same antijoin using NOT EXISTS.

CHAPTER 05

SQL in practice

Schema changes, transactions, views, CTEs and analytical queries

01 / Define, change and query safely

DDL defines schemas with CREATE, ALTER and DROP; DML changes rows with INSERT, UPDATE and DELETE.

ACID summarizes atomicity (all or nothing), consistency (preserved declared rules and application invariants), isolation (controlled interference) and durability (committed changes survive failures according to system guarantees).

Bind user data as parameters instead of concatenating it into SQL text.

Check affected rows and constraints.

Watch the trap: A parameter is a data value, not a fragment of SQL syntax.

Retrieve by doing: Create a table, insert a row, roll back the insert, and verify zero rows.

02 / Views, CTEs and reusable query structure

A view names a stored query.

A common table expression introduced with WITH names a query result for the duration of one statement.

Set operators combine compatible result schemas.

Recursive CTEs have an anchor query and a recursive step and can express hierarchy traversal.

Watch the trap: A CTE is a query organization tool, not a guarantee of a temporary table.

Retrieve by doing: Use a CTE to find students with at least two enrolments.

03 / Window functions and an analytical capstone

A window function calculates across related rows while retaining the original row grain.

ROW_NUMBER assigns a sequence, RANK leaves gaps after ties, and DENSE_RANK does not.

For a running total, state the frame explicitly: ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW.

Capstone method: write the output grain in words, join only compatible facts, handle NULL deliberately, aggregate, then apply the window calculation.

Watch the trap: Window calculations do not automatically reduce the number of rows.

Retrieve by doing: Show each sale with its city total and share of that total.

CHAPTER 06

Dependencies and keys

Attribute closure, candidate keys, covers and decomposition tests

01 / Functional dependencies and attribute closure

A functional dependency X -> Y says any two legal tuples agreeing on all attributes of X must agree on all attributes of Y.

Armstrong's rules are reflexivity, augmentation and transitivity.

Attribute closure X+ is the set determined by X under F.

For R(A,B,C,D), F={A->B,B->C,AC->D}, closure A begins {A}, adds B, then C, then D.

Watch the trap: An FD can hold in a sample by coincidence. Keys are derived from the schema's declared dependencies.

Retrieve by doing: Compute B+ for the same dependencies.

02 / Find all candidate keys and a minimal cover

A candidate key is an inclusion-minimal superkey, not necessarily the shortest key among all keys.

To enumerate keys, attributes absent from every nontrivial FD right side must be in every key.

A minimal cover is an equivalent FD set with singleton right sides, no extraneous left-side attributes and no redundant dependencies.

Minimal covers need not be unique.

Watch the trap: Prime means membership in any candidate key, not just in the primary key.

Retrieve by doing: Find keys of ABC with only A->B.

03 / Lossless joins and dependency preservation

Decomposition replaces R with smaller relation schemas whose union of attributes is R.

For a binary decomposition R1,R2 under functional dependencies, the lossless criterion is that the common attributes determine all of R1 or all of R2 in F+.

Dependency preservation asks whether enforcing dependencies projected onto the individual components suffices to enforce all original FDs without joining.

For decompositions into more than two relations, use a chase/tableau or a valid sequence of lossless binary steps.

Watch the trap: A smaller schema is not automatically a correct decomposition. Check losslessness before celebrating reduced redundancy.

Retrieve by doing: Show that AB and BC rejoin losslessly when B->C holds in {(a1,b,c),(a2,b,c)}.

CHAPTER 07

Normal forms

Remove anomalies with 1NF, 2NF, 3NF, BCNF and higher-form context

01 / Anomalies, 1NF and 2NF

Redundancy creates update, insertion and deletion anomalies.

First normal form requires atomic attribute values relative to the chosen model and no repeating groups.

Second normal form requires 1NF and no nonprime attribute partially dependent on a proper subset of any candidate key.

A single-attribute candidate key has no nonempty proper subset, so partial dependencies of the usual kind do not arise through that key.

Watch the trap: Normalization concerns dependencies and legal instances, not the visual neatness of a current table.

Retrieve by doing: Return course titles and counts, including courses with no students.

02 / 3NF and BCNF: use the exact conditions

A schema is in third normal form if for every nontrivial FD X->A in F+, either X is a superkey or A is prime.

Boyce-Codd normal form requires every nontrivial FD determinant to be a superkey.

To decompose a BCNF violation X->Y, use R1=X union Y and R2=R-(Y-X).

3NF synthesis starts from a minimal cover, creates a relation for each determinant and its dependents, removes contained schemas, and adds a relation containing a candidate key if none already contains one.

Watch the trap: A prime right-side attribute can save 3NF; it cannot save BCNF.

Retrieve by doing: For A->B and B->C on ABC, test whether B is a superkey.

03 / Multivalued dependencies and higher forms

An MVD X ->> Y states that, for a fixed X, Y values vary independently of the remaining attributes Z.

An MVD is trivial when Y is contained in X or X union Y is the full schema.

A join dependency says a relation equals the join of specified projections.

The official syllabus says "normal form" without listing levels.

Watch the trap: Two multivalued attributes are not automatically independent; the MVD must be part of the semantics.

Retrieve by doing: Compare storage for 5 independent hobbies and 7 languages.

CHAPTER 08

Files and indexes

Pages, records, ordered files, hashing and B+ tree arithmetic

01 / Page organization and I/O cost

A database moves data between storage and memory in blocks or pages.

Unspanned records fit entirely within one block; spanned records may cross block boundaries and require continuation information.

A heap file has no key order: insertion is convenient but an unindexed search may scan all b blocks.

For range queries, ordered/clustered storage can retrieve adjacent matching records with fewer page reads.

Watch the trap: Do not compute ceil(N*R/B) for unspanned fixed records; unused space in each block matters.

Retrieve by doing: Recalculate for usable block size 1000, record size 300 and 10 records.

02 / Dense, sparse and secondary indexes

An index stores search-key values with pointers so the DBMS can locate records without scanning the whole file.

A sparse index stores entries for only some keys, commonly one per data block.

Traditional primary indexes refer to an ordered unique key; clustering indexes refer to an ordering nonkey field; secondary indexes refer to a nonordering field.

A multilevel index recursively indexes the previous level until the top fits in one block.

Watch the trap: Index levels and total search I/Os differ if the root is cached or a data block must also be read.

Retrieve by doing: Repeat with 1,000,000 entries and 100 entries per index block.

03 / B+ trees, hashing and plan selection

A B+ tree is a balanced multilevel search index.

If an internal page fits p pointers of size P and p-1 keys of size K, solve pP+(p-1)K<=B after subtracting any header.

Insertion may split a full leaf, copy a separator to the parent, and propagate internal splits.

Hash indexes direct equality searches to buckets and handle overflow or resize by a stated method.

Watch the trap: Do not reuse an internal-node fanout formula for leaf capacity.

Retrieve by doing: Find the internal fanout for B=1000, K=10, P=6 with no header.

CHAPTER 09

Data types and cleaning

Measurement scales, missing values, encoding and leakage

01 / Know what a value means

Nominal values are categories without intrinsic order, such as city.

Discrete variables take countable values, while continuous measurements are conceptually real-valued even if stored at finite precision.

A database type is an implementation domain: INTEGER, DECIMAL, floating-point, text, date/time, Boolean and binary objects.

Before transforming a column, record its units, valid range, missing-value representation and measurement scale.

Watch the trap: Numeric storage does not imply a ratio-scale quantity.

Retrieve by doing: One-hot encode Delhi and Pune for [Delhi,Pune,Delhi].

02 / Missing values, duplicates and outliers

Profile data before repairing it: row counts, key uniqueness, missing fractions, category frequencies and numeric ranges.

Missingness may be completely random, dependent on observed values, or dependent on missing values themselves.

Use a median for a robust central-value imputation when appropriate; use group-specific values only if the grouping is meaningful and sufficiently populated.

An outlier may be an error or a real extreme event.

Watch the trap: Do not fit an imputer on the test set or silently treat every extreme value as an error.

Retrieve by doing: Apply the same fitted median 11.5 to a separate test list [None,20].

03 / Encoding, integration and reproducible pipelines

One-hot encoding creates one indicator per category.

Data integration aligns schemas, units, keys and meanings across sources.

A reproducible pipeline separates fit from transform, versions its schema and parameters, and checks output invariants.

Practical mini-project: integrate students and enrolments, report unmatched foreign keys, derive course completion counts, encode cities and save the mapping.

Watch the trap: Never refit category-code assignments independently on training and test data.

Retrieve by doing: Convert [100,250,0] centimetres to metres and assert all values are nonnegative.

CHAPTER 10

Normalize and discretize

Scaling formulas, bin boundaries and the meaning of transformed data

01 / Min-max, z-score and decimal scaling

Numeric normalization rescales values; it does not normalize a relational schema.

Z-score standardization is z=(x-mean)/standard_deviation.

Decimal scaling divides by 10^j, where j is the smallest integer making the maximum absolute transformed value less than 1 under the strict convention.

Fit parameters on training data and reuse them.

Watch the trap: Standardization does not make an arbitrary distribution Gaussian.

Retrieve by doing: Scale 40 using fitted training min=10 and max=30.

02 / Equal-width and equal-frequency bins

Discretization replaces a continuous value with a category or interval.

Equal-frequency (quantile) bins aim for similar numbers of observations, not equal numeric widths.

Binning can reduce sensitivity to noise and improve interpretability but loses within-bin information.

For exam problems, first sort if needed, compute boundaries or bucket sizes, then apply the declared endpoint rule to every value.

Watch the trap: Equal-frequency does not mean equal-width, and ties make quantile bins ambiguous.

Retrieve by doing: Use two equal-width bins on [0,100] for 0,49,50,100.

03 / Smoothing, transformations and information loss

Binning by means replaces every value in a bin by its bin mean; bin medians use the median; bin boundaries use the nearest boundary with a stated tie rule.

Log transforms compress large positive magnitudes. log(0) is undefined; log1p(x)=log(1+x) permits zero for x>=0 but changes the interpretation.

Aggregation reduces granularity: transactions become daily totals, and individual variation within a day disappears.

Choose transformations by purpose and verify properties.

Watch the trap: Smoothing loses information; it is not an invertible rescaling.

Retrieve by doing: Apply log1p to 0,9,99 and round to three decimals.

CHAPTER 11

Sampling and compression

Representative samples, streaming selection and compact representations

01 / Random, stratified, cluster and systematic samples

Simple random sampling without replacement gives every fixed-size subset the same probability; with replacement, the same unit may appear more than once.

Stratified sampling divides the population into meaningful groups and samples within every group.

Cluster sampling selects groups and then all or some units within selected groups.

Sampling reduces computation but introduces sampling error and can miss rare patterns.

Watch the trap: A fixed seed provides reproducibility, not freedom from bias.

Retrieve by doing: Select two units from each of two strata using a seeded generator.

02 / Reservoir sampling and estimator checks

Reservoir sampling maintains a uniform size-k sample from a stream of unknown final length using O(k) storage.

The incoming ith item is selected with probability k/i.

When estimating a mean from a simple random sample, the sample mean is unbiased under the sampling design.

Use simulation to check an implementation over many runs, but distinguish evidence from proof.

Watch the trap: Always selecting a replacement slot among k would force every new item into the sample and bias it.

Retrieve by doing: Compute the marginal inclusion probability for k=5 from a stream of 100 items.

03 / Lossless and lossy compression

Lossless compression allows exact reconstruction; lossy compression discards some information to reduce size.

Run-length encoding stores repeated consecutive symbols as (symbol,count), so it benefits long runs but can expand alternating data.

Huffman coding builds a prefix-free variable-length code by repeatedly merging the two least frequent symbols.

Quantization is lossy: nearby values are mapped to one representative.

Watch the trap: RLE size depends on runs, not merely the total number of repeated symbols.

Retrieve by doing: Delta-encode [100,103,109,110], keeping the first value.

CHAPTER 12

Warehousing and OLAP

Grain, star schemas, hierarchies, cubes and correct measure roll-ups

01 / Choose the fact grain and warehouse schema

A warehouse integrates historical data for analysis.

Declare the fact grain before selecting measures: one row per sale line, per store-product-day snapshot, or per completed process.

A star schema has a central fact table and comparatively denormalized dimensions.

Use surrogate dimension keys when integrating source identities or tracking history.

Watch the trap: Joining fact tables directly can multiply facts. Aggregate to a compatible grain or use a carefully designed model.

Retrieve by doing: Roll the same facts up to product totals.

02 / Concept hierarchies, cubes and OLAP operations

A concept hierarchy links detail to broader categories: city -> state -> country -> ALL, or day -> month -> quarter -> year -> ALL.

Roll-up aggregates to a coarser level; drill-down moves to finer detail if that detail is available.

A cuboid is an aggregation at a chosen combination of dimension levels.

A full cube may be expensive: the number of possible cells is a product of dimension cardinalities, and many cells are empty.

Watch the trap: Do not multiply cardinalities of cities or products when the question asks for the number of hierarchy-level cuboids.

Retrieve by doing: Count cuboids for dimensions with 4,3,2 levels including ALL.

03 / Measures: additive, algebraic and holistic

Additive measures can be summed across all relevant dimensions, such as transaction revenue.

A separate classification concerns how aggregates can be combined.

Holistic aggregates such as an exact median generally need an amount of state that grows with the data; medians of groups are insufficient.

For a weighted average selling price, sum revenue and units separately, then divide.

Watch the trap: An average of averages is valid only with appropriate equal weights or explicit count weighting.

Retrieve by doing: Compute the overall mean of groups (mean=10,count=2) and (mean=30,count=8).

Authentic exam practice

Open the exact original question and all its options before answering. The site has a 20-question mixed set and a full bank of 22 database/warehouse PYQs. Original conceptual questions are labelled separately.

GATE DA 2024

Q26 / page 16 / MCQ / Set containment
Q27 / page 17 / MCQ / Z-score scaling
Q31 / page 19 / NAT / Filtered join
Q46 / page 29 / MSQ / Dependency closure
Q55 / page 37 / MSQ / Index selection
Original paper

Official key

GATE DA 2025

Q16 / page 11 / MCQ / Dependency preservation
Q17 / page 12 / MCQ / Algebra interpretation
Q33 / page 18 / NAT / Scalar subquery
Q56 / page 30 / MSQ / Keys and NULL
Q57 / page 31 / MSQ / Keys and BCNF
Q62 / page 32 / NAT / Relational division
Original paper

Official key

GATE DA 2026

Q17 / page 12 / MCQ / Candidate keys
Q18 / page 12 / MCQ / OLAP navigation
Q26 / page 18 / MSQ / Referential integrity
Q32 / page 21 / NAT / Index capacity
Q41 / page 26 / MCQ / B+ insertion
Q42 / page 27 / MCQ / Natural joins
Q43 / page 28 / MCQ / Cuboid counting
Q51 / page 34 / MSQ / Grouped SQL
Q59 / page 38 / NAT / Tuple calculus
Q60 / page 38 / NAT / Correlated counts
Q61 / page 39 / NAT / ER mapping
Original paper

Official key

2026 Q59: the official key accepts the range 2 to 3. The normal set-semantics reasoning yields 2; the site preserves the official accepted range.